

LABSHEET-9

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ROLL NO-20

SEC-B

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from google.colab import files
pd.set_option('display.max_columns', None)
sns.set_style("whitegrid")
print("Libraries imported successfully!")
```

Libraries imported successfully!

1. Select a real-world dataset such as the Global Superstore Sales Dataset, Netflix Dataset, World Happiness Report, or COVID-19 Dataset.

```
uploaded = files.upload()
```

Choose files netflix\_titles.csv  
netflix\_titles.csv(text/csv) - 3399671 bytes, last modified: 21/09/2026 - 100% done  
Saving netflix\_titles.csv to netflix\_titles (1).csv

```
df = pd.read_csv("netflix_titles.csv")
print("Dataset loaded successfully!")
print("Shape:", df.shape)
df.head()
```

Dataset loaded successfully!  
Shape: (8807, 12)

|   | show_id | type    | title                | director        | cast                                              | country       | date_added         | release_year | rating | duration  | listed_in                                       | description                                       |
|---|---------|---------|----------------------|-----------------|---------------------------------------------------|---------------|--------------------|--------------|--------|-----------|-------------------------------------------------|---------------------------------------------------|
| 0 | s1      | Movie   | Dick Johnson Is Dead | Kirsten Johnson | NaN                                               | United States | September 25, 2021 | 2020         | PG-13  | 90 min    | Documentaries                                   | As her father nears the end of his life, filmm... |
| 1 | s2      | TV Show | Blood & Water        | NaN             | Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban... | South Africa  | September 24, 2021 | 2021         | TV-MA  | 2 Seasons | International TV Shows, TV Dramas, TV Mysteries | After crossing paths at a party, a Cape Town t... |
| 2 | s3      | TV Show | Ganglands            | Julien          | Sami Bouajila, Tracy Gotoas,                      | NaN           | September 24, 2021 | 2021         | TV-MA  | 1 Season  | Crime TV Shows, International                   | To protect his family from a                      |

Next steps: [Generate code with df](#) [New interactive sheet](#)

2. Analyze the dataset to identify the primary business problem or analytical objective.

```
print("""
BUSINESS PROBLEM / ANALYTICAL OBJECTIVE
The objective of this analysis is to understand the content available
on Netflix and identify important patterns related to:
1. Movies and TV Shows
2. Content released over the years
3. Popular genres/categories
4. Countries producing Netflix content
5. Content ratings
6. Content duration
7. Recent content trends
These insights can help understand Netflix's content strategy and
```

```
identify areas of growth and improvement.
""")
```

#### BUSINESS PROBLEM / ANALYTICAL OBJECTIVE

The objective of this analysis is to understand the content available on Netflix and identify important patterns related to:

1. Movies and TV Shows
2. Content released over the years
3. Popular genres/categories
4. Countries producing Netflix content
5. Content ratings
6. Content duration
7. Recent content trends

These insights can help understand Netflix's content strategy and identify areas of growth and improvement.

### 3. Perform exploratory data analysis to discover significant trends, patterns, correlations, and anomalies.

#### Cell 4 — Basic dataset information

```
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
print("\nColumn Names:")
print(df.columns.tolist())
print("\nDataset Information:")
df.info()
```

#### Cell 5 — Statistical summary

```
df.describe(include='all').T
```

|              | count  | unique | top                                               | freq | mean        | std      | min    | 25%    | 50%    | 75%    | max    |
|--------------|--------|--------|---------------------------------------------------|------|-------------|----------|--------|--------|--------|--------|--------|
| show_id      | 8807   | 8807   | s8807                                             | 1    | NaN         | NaN      | NaN    | NaN    | NaN    | NaN    | NaN    |
| type         | 8807   | 2      | Movie                                             | 6131 | NaN         | NaN      | NaN    | NaN    | NaN    | NaN    | NaN    |
| title        | 8807   | 8807   | Zubaan                                            | 1    | NaN         | NaN      | NaN    | NaN    | NaN    | NaN    | NaN    |
| director     | 6173   | 4528   | Rajiv Chilaka                                     | 19   | NaN         | NaN      | NaN    | NaN    | NaN    | NaN    | NaN    |
| cast         | 7982   | 7692   | David Attenborough                                | 19   | NaN         | NaN      | NaN    | NaN    | NaN    | NaN    | NaN    |
| country      | 7976   | 748    | United States                                     | 2818 | NaN         | NaN      | NaN    | NaN    | NaN    | NaN    | NaN    |
| date_added   | 8797   | 1767   | January 1, 2020                                   | 109  | NaN         | NaN      | NaN    | NaN    | NaN    | NaN    | NaN    |
| release_year | 8807.0 | NaN    | NaN                                               | NaN  | 2014.180198 | 8.819312 | 1925.0 | 2013.0 | 2017.0 | 2019.0 | 2021.0 |
| rating       | 8803   | 17     | TV-MA                                             | 3207 | NaN         | NaN      | NaN    | NaN    | NaN    | NaN    | NaN    |
| duration     | 8804   | 220    | 1 Season                                          | 1793 | NaN         | NaN      | NaN    | NaN    | NaN    | NaN    | NaN    |
| listed_in    | 8807   | 514    | Dramas, International Movies                      | 362  | NaN         | NaN      | NaN    | NaN    | NaN    | NaN    | NaN    |
| description  | 8807   | 8775   | Paranormal activity at a lush, abandoned prope... | 4    | NaN         | NaN      | NaN    | NaN    | NaN    | NaN    | NaN    |

#### Cell 6 — Check missing values

```
missing_values = df.isnull().sum()
missing_percentage = (df.isnull().sum() / len(df)) * 100
missing_df = pd.DataFrame({
    "Missing Values": missing_values,
    "Percentage": missing_percentage
})
missing_df = missing_df.sort_values(
    by="Percentage",
    ascending=False
)
missing_df
```

|              | Missing Values | Percentage |   |
|--------------|----------------|------------|-----------------------------------------------------------------------------------|
| director     | 2634           | 29.908028  |  |
| country      | 831            | 9.435676   |                                                                                   |
| cast         | 825            | 9.367549   |                                                                                   |
| date_added   | 10             | 0.113546   |                                                                                   |
| rating       | 4              | 0.045418   |                                                                                   |
| duration     | 3              | 0.034064   |                                                                                   |
| show_id      | 0              | 0.000000   |                                                                                   |
| type         | 0              | 0.000000   |                                                                                   |
| title        | 0              | 0.000000   |                                                                                   |
| release_year | 0              | 0.000000   |                                                                                   |
| listed_in    | 0              | 0.000000   |                                                                                   |
| description  | 0              | 0.000000   |                                                                                   |

Next steps:

[Generate code with missing\\_df](#)[New interactive sheet](#)**Cell 7 — Check duplicate records**

```
print("Number of duplicate rows:", df.duplicated().sum())
```

Number of duplicate rows: 0

```
df = df.drop_duplicates()
print("Shape after removing duplicates:", df.shape)
```

Shape after removing duplicates: (8807, 12)

**4. Develop a set of meaningful visualizations such as bar charts, line graphs, heatmaps, and dashboards to support the analysis.****Cell 8 — Handle missing values**

```
text_columns = [
    'director',
    'cast',
    'country',
    'rating',
    'duration',
    'listed_in',
    'description'
]
for col in text_columns:
    if col in df.columns:
        df[col] = df[col].fillna("Unknown")
print("Missing text values handled.")
```

Missing text values handled.

**Cell 9 — Convert date column**

```
df['date_added'] = pd.to_datetime(
    df['date_added'],
    errors='coerce'
)
print("Date column converted successfully.")
df[['date_added']].head()
```

Date column converted successfully.

|   | date_added |  |
|---|------------|--|
| 0 | 2021-09-25 |  |
| 1 | 2021-09-24 |  |
| 2 | 2021-09-24 |  |
| 3 | 2021-09-24 |  |
| 4 | 2021-09-24 |  |

Cell 10 — Create useful year columns

```
df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month
df[['date_added', 'year_added', 'month_added']].head()
```

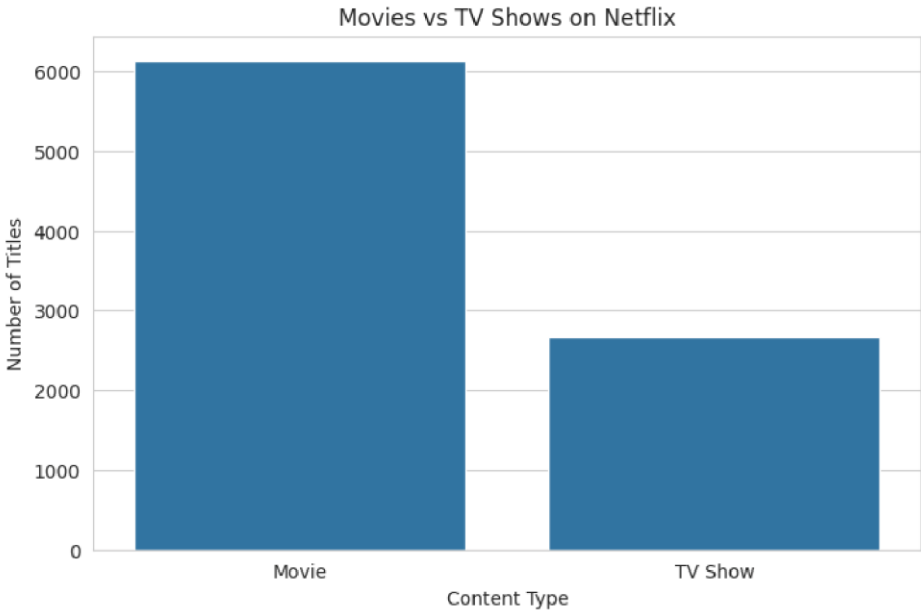
|   | date_added | year_added | month_added |  |
|---|------------|------------|-------------|--|
| 0 | 2021-09-25 | 2021.0     | 9.0         |  |
| 1 | 2021-09-24 | 2021.0     | 9.0         |  |
| 2 | 2021-09-24 | 2021.0     | 9.0         |  |
| 3 | 2021-09-24 | 2021.0     | 9.0         |  |
| 4 | 2021-09-24 | 2021.0     | 9.0         |  |

5. Identify Key Performance Indicators (KPIs) and highlight important business metrics relevant to the selected dataset.

Visualization 1 — Movies vs TV Shows

Cell 11

```
content_type = df['type'].value_counts()
plt.figure(figsize=(8, 5))
sns.barplot(
    x=content_type.index,
    y=content_type.values
)
plt.title("Movies vs TV Shows on Netflix")
plt.xlabel("Content Type")
plt.ylabel("Number of Titles")
plt.show()
```



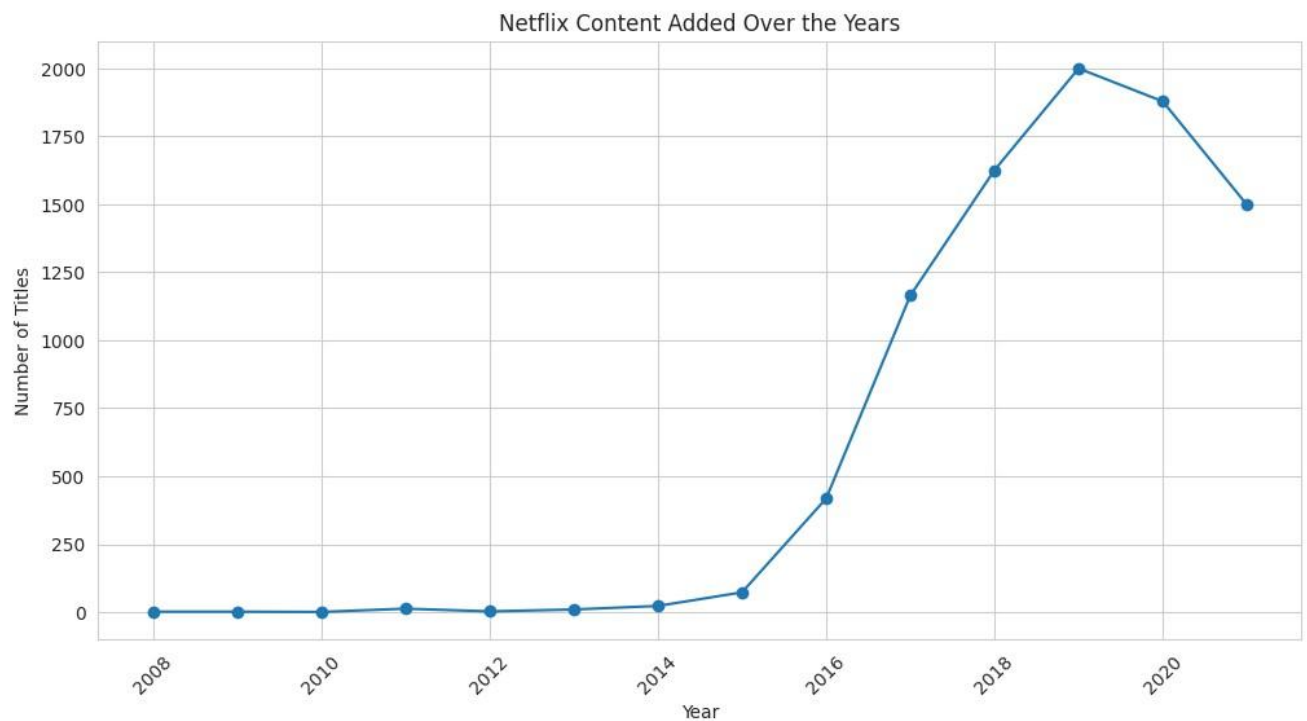
Visualization 2 — Content Added Over the Years

Cell 12

```

yearly_content = df['year_added'].value_counts().sort_index()
plt.figure(figsize=(12, 6))
plt.plot(
    yearly_content.index,
    yearly_content.values,
    marker='o'
)
plt.title("Netflix Content Added Over the Years")
plt.xlabel("Year")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()

```



### Visualization 3 — Top Genres

The `listed_in` column contains multiple genres, so we need to separate them.

#### Cell 13

```

genres = df['listed_in'].dropna().str.split(',')
genre_list = []
for genre in genres:
    genre_list.extend(genre)
genre_counts = pd.Series(genre_list).value_counts()
genre_counts.head(10)

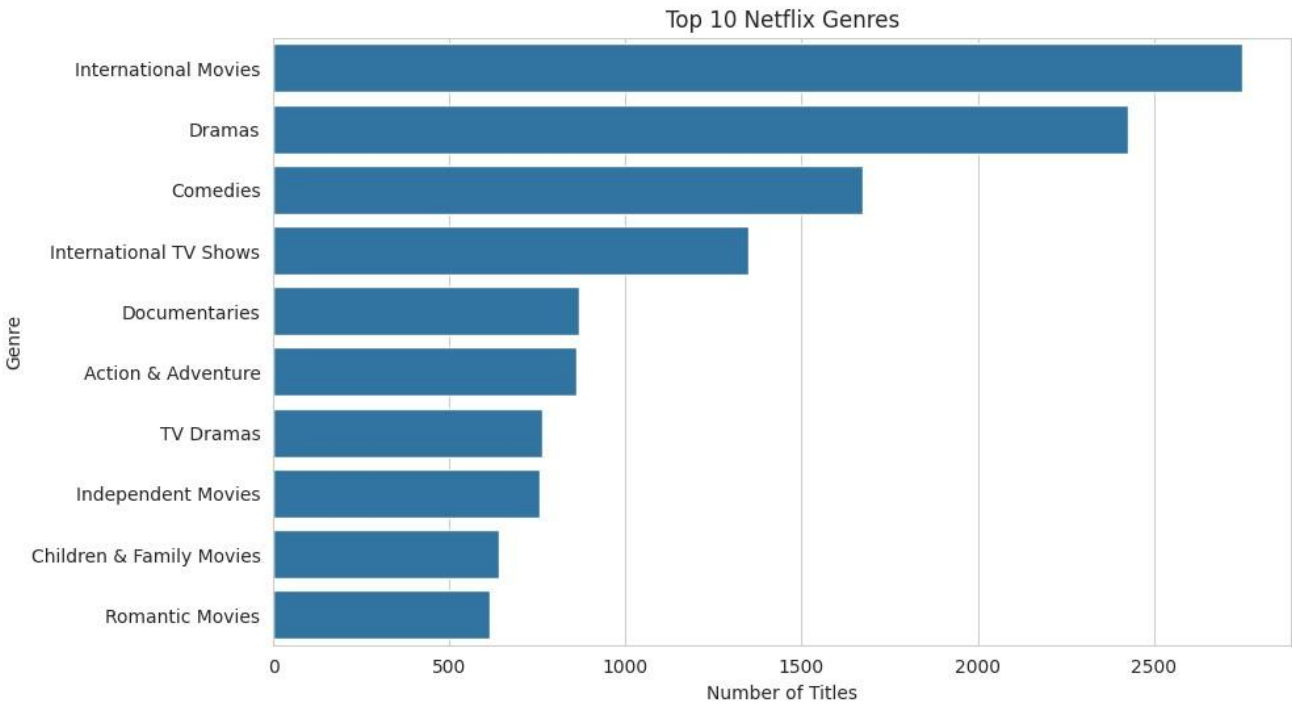
```

|                          | count |
|--------------------------|-------|
| International Movies     | 2752  |
| Dramas                   | 2427  |
| Comedies                 | 1674  |
| International TV Shows   | 1351  |
| Documentaries            | 869   |
| Action & Adventure       | 859   |
| TV Dramas                | 763   |
| Independent Movies       | 756   |
| Children & Family Movies | 641   |
| Romantic Movies          | 616   |

dtype: int64

Cell 14 — Plot top 10 genres

```
top_genres = genre_counts.head(10)
plt.figure(figsize=(10, 6))
sns.barplot(
    x=top_genres.values,
    y=top_genres.index
)
plt.title("Top 10 Netflix Genres")
plt.xlabel("Number of Titles")
plt.ylabel("Genre")
plt.show()
```



Visualization 4 — Content by Country  
Cell 15

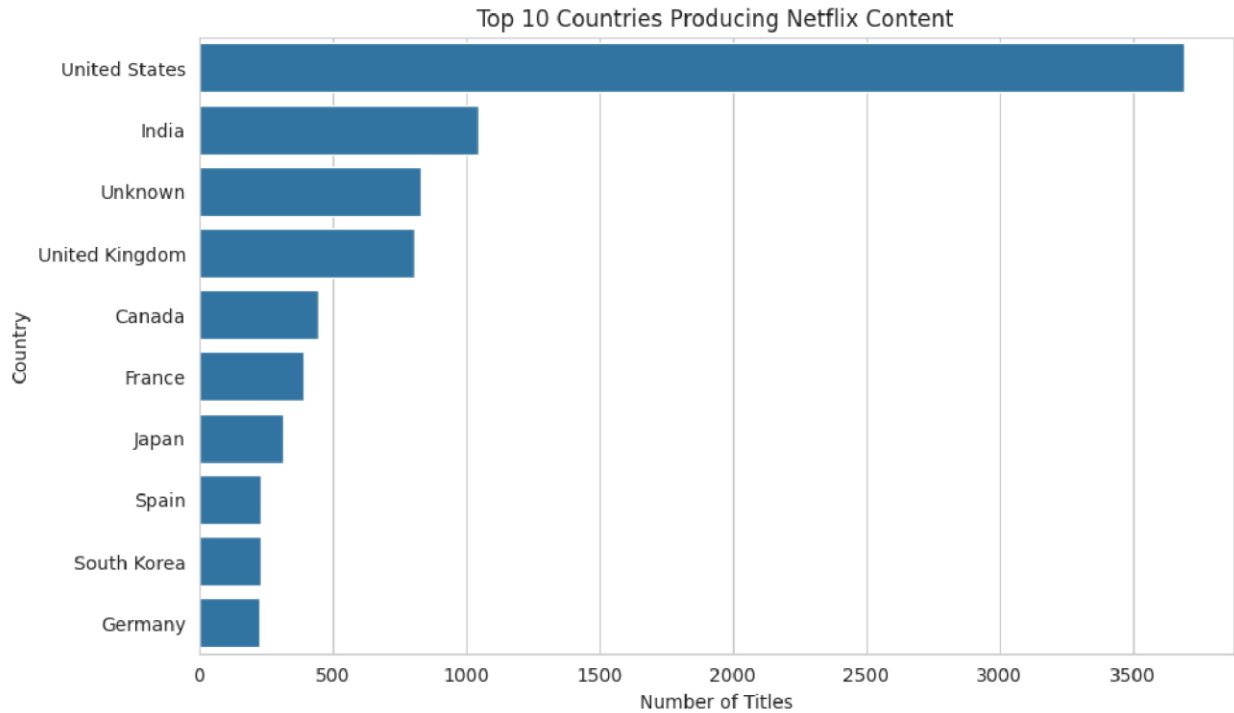
```
countries = df['country'].dropna().str.split(', ')
country_list = []
for country in countries:
    country_list.extend(country)
country_counts = pd.Series(country_list).value_counts()
country_counts.head(10)
```

|                | count |
|----------------|-------|
| United States  | 3689  |
| India          | 1046  |
| Unknown        | 831   |
| United Kingdom | 804   |
| Canada         | 445   |
| France         | 393   |
| Japan          | 318   |
| Spain          | 232   |
| South Korea    | 231   |
| Germany        | 226   |

dtype: int64

Cell 16

```
top_countries = country_counts.head(10)
plt.figure(figsize=(10, 6))
sns.barplot(
    x=top_countries.values,
    y=top_countries.index
)
plt.title("Top 10 Countries Producing Netflix Content")
plt.xlabel("Number of Titles")
plt.ylabel("Country")
plt.show()
```



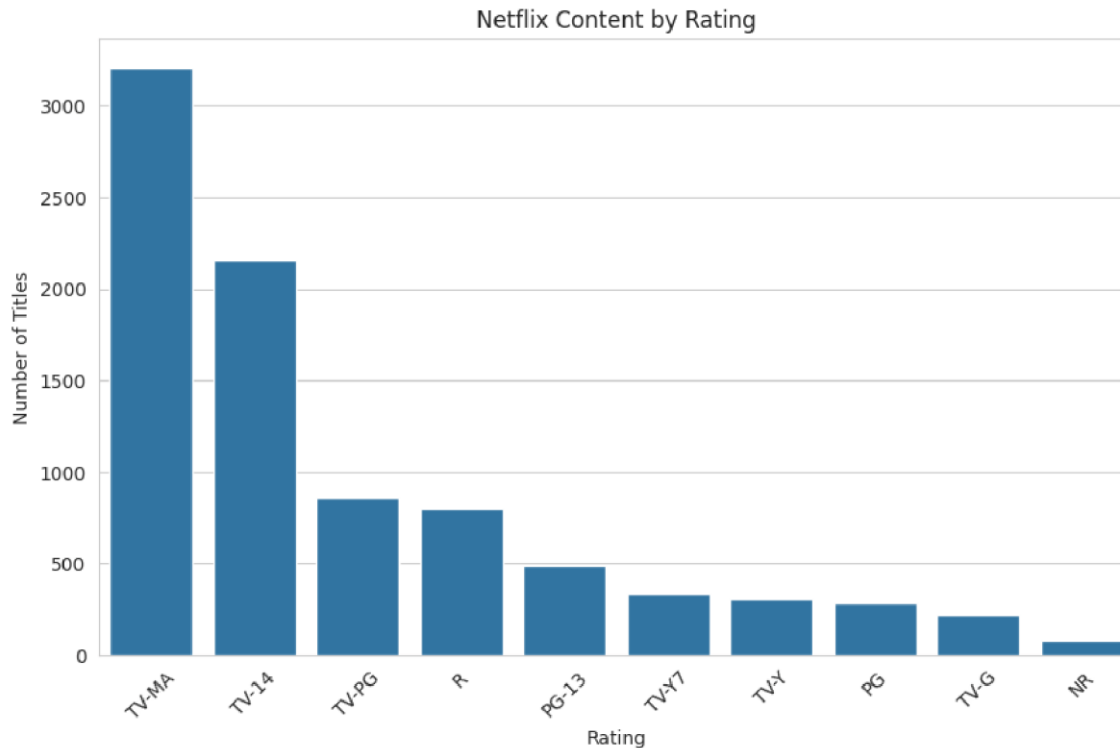
Visualization 5 — Content Rating Distribution

Cell 17

```

rating_counts = df['rating'].value_counts().head(10)
plt.figure(figsize=(10, 6))
sns.barplot(
    x=rating_counts.index,
    y=rating_counts.values
)
plt.title("Netflix Content by Rating")
plt.xlabel("Rating")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()

```



### Visualization 6 — Movies vs TV Shows by Year

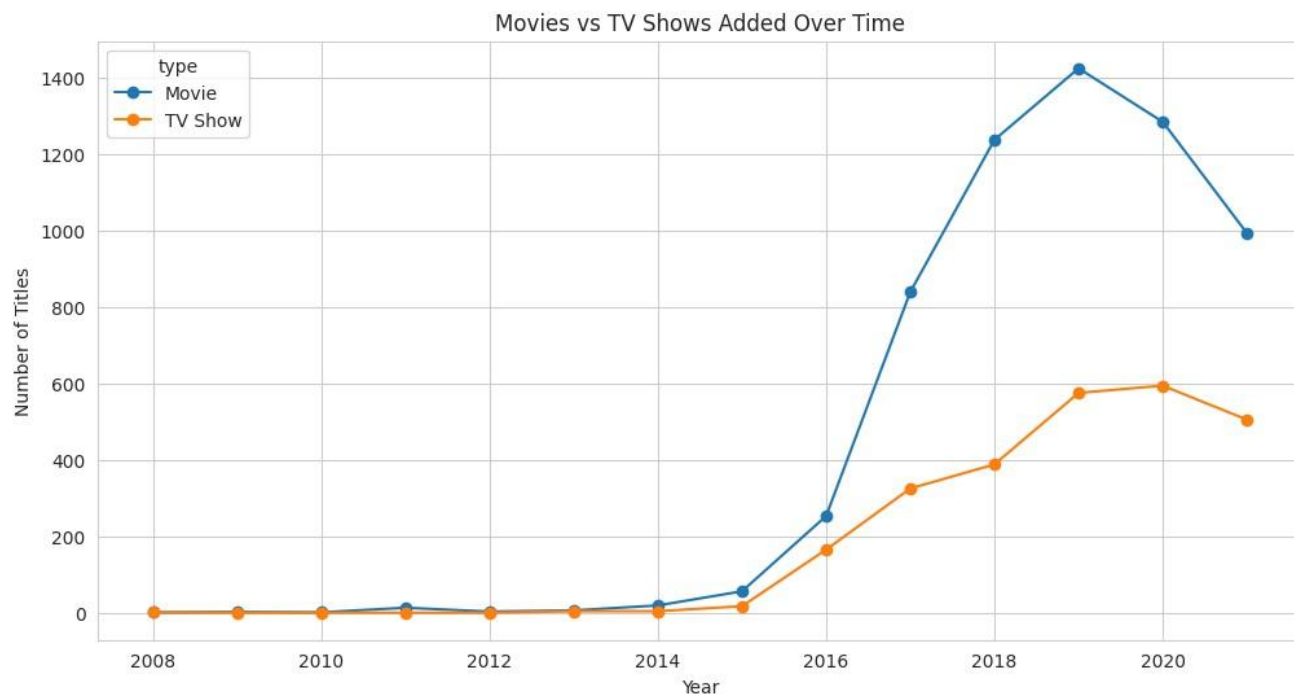
Cell 18

```

year_type = pd.crosstab(
    df['year_added'],
    df['type']
)
plt.figure(figsize=(12, 6))
year_type.plot(
    kind='line',
    marker='o',
    figsize=(12, 6)
)
plt.title("Movies vs TV Shows Added Over Time")
plt.xlabel("Year")
plt.ylabel("Number of Titles")
plt.grid(True)
plt.show()

```

&lt;Figure size 1200x600 with 0 Axes&gt;

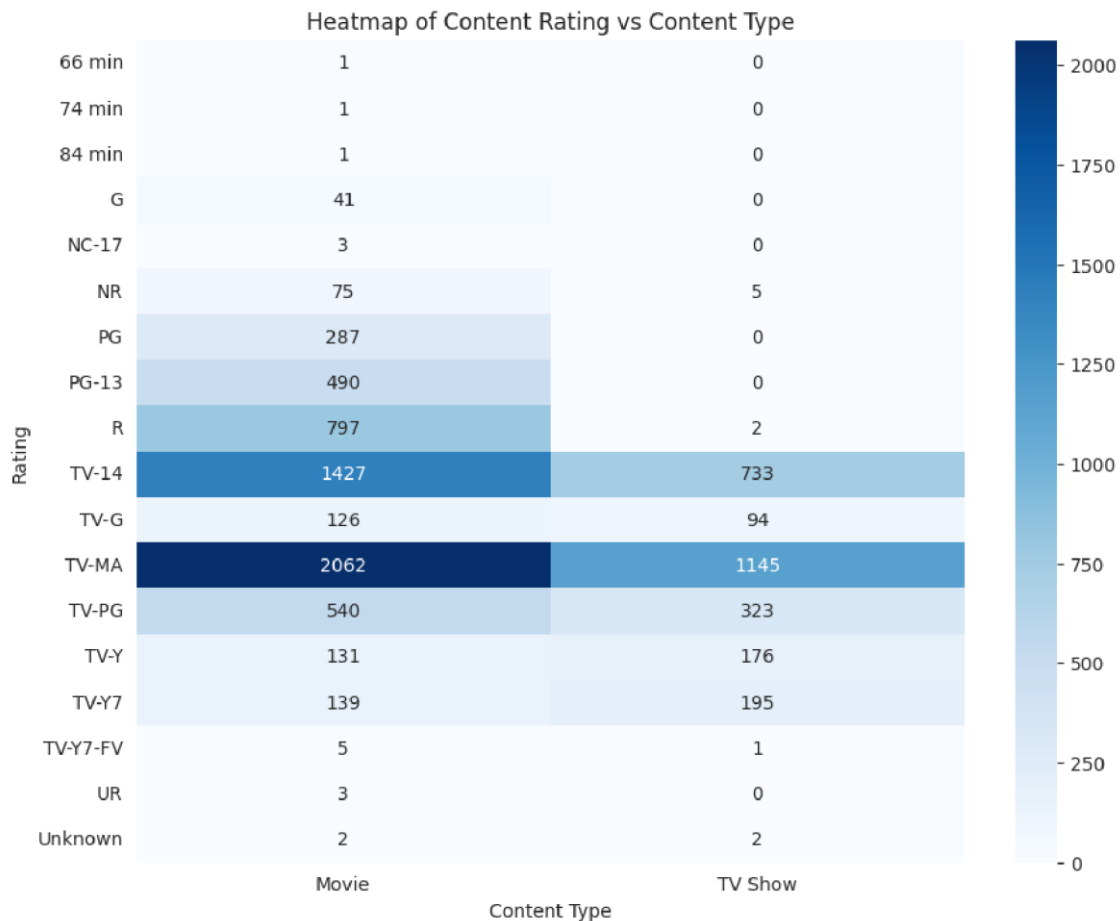


### Visualization 7 — Heatmap

The heatmap helps identify relationships between content type and rating.

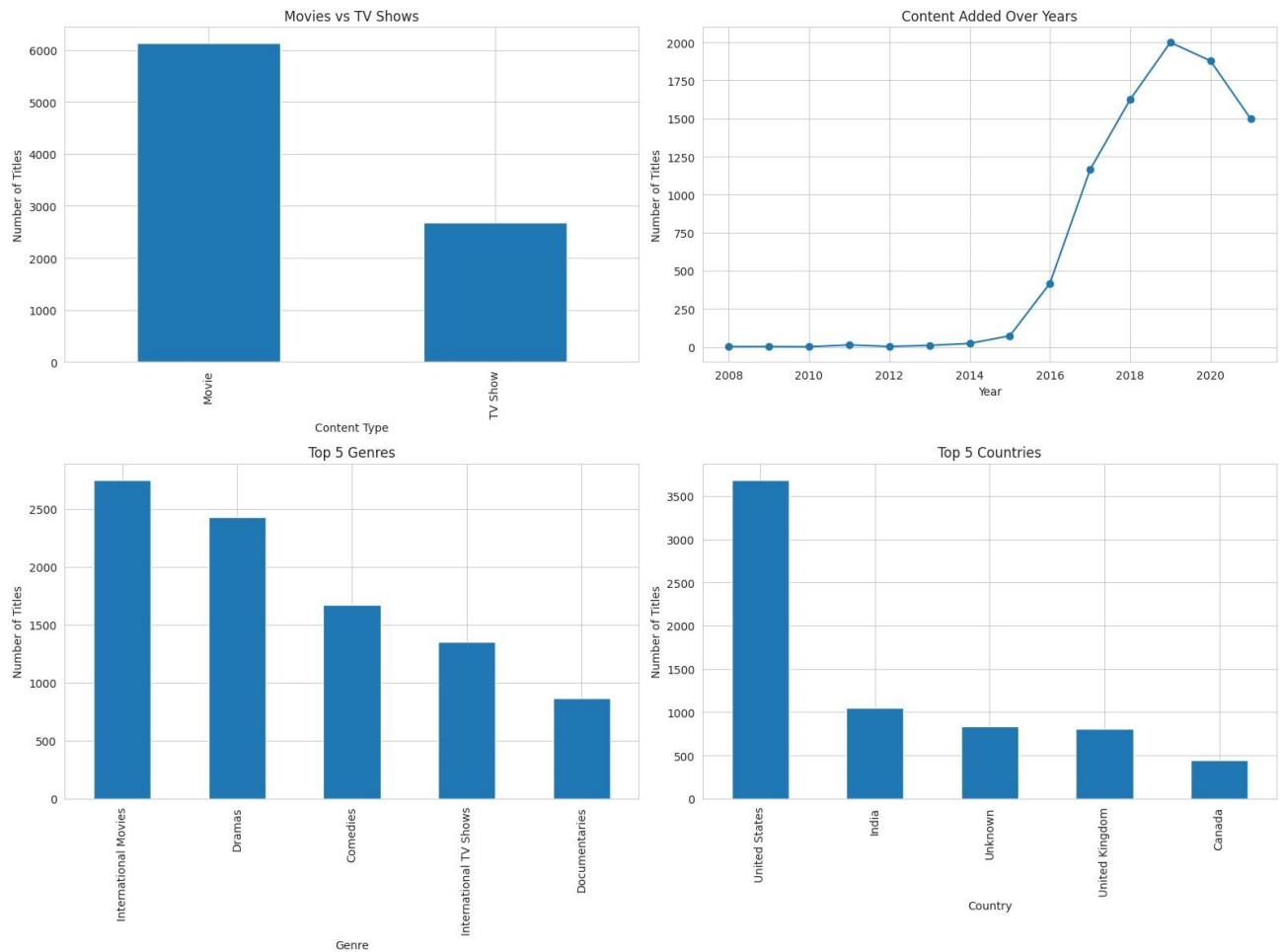
#### Cell 19

```
rating_type = pd.crosstab(
    df['rating'],
    df['type']
)
plt.figure(figsize=(10, 8))
sns.heatmap(
    rating_type,
    annot=True,
    fmt='d',
    cmap='Blues'
)
plt.title("Heatmap of Content Rating vs Content Type")
plt.xlabel("Content Type")
plt.ylabel("Rating")
plt.show()
```



6. Arrange the visualizations in a logical sequence to create a coherent data story that explains the problem, analysis, findings, and recommendations.

```
fig, axes = plt.subplots(2, 2, figsize=(16, 12))
content_type.plot(
    kind='bar',
    ax=axes[0, 0]
)
axes[0, 0].set_title("Movies vs TV Shows")
axes[0, 0].set_xlabel("Content Type")
axes[0, 0].set_ylabel("Number of Titles")
yearly_content.plot(
    kind='line',
    marker='o',
    ax=axes[0, 1]
)
axes[0, 1].set_title("Content Added Over Years")
axes[0, 1].set_xlabel("Year")
axes[0, 1].set_ylabel("Number of Titles")
top_genres.head(5).plot(
    kind='bar',
    ax=axes[1, 0]
)
axes[1, 0].set_title("Top 5 Genres")
axes[1, 0].set_xlabel("Genre")
axes[1, 0].set_ylabel("Number of Titles")
top_countries.head(5).plot(
    kind='bar',
    ax=axes[1, 1]
)
axes[1, 1].set_title("Top 5 Countries")
axes[1, 1].set_xlabel("Country")
axes[1, 1].set_ylabel("Number of Titles")
plt.tight_layout()
plt.show()
```



## 8. Prepare actionable recommendations based on the analytical findings to support business decision-making.

### Cell 22

```
print("===== BUSINESS INSIGHTS =====\n")
print("1. Content Mix:")
print(
    f"Netflix contains {total_movies} Movies and "
    f"{total_tv_shows} TV Shows in the dataset."
)
print("\n2. Most Common Genres:")
print(
    "The most common genres are:"
)
print(top_genres.head(5))
print("\n3. Leading Content-Producing Countries:")
print(
    "The leading countries by number of titles are:"
)
print(top_countries.head(5))
```

```

print("\n4. Content Growth:")
print(
    "The yearly content trend shows how Netflix's content "
    "library has changed over time."
)
print("\n5. Content Ratings:")
print(
    "The rating distribution shows which audience categories "
    "are most represented in the dataset."
)

```

===== BUSINESS INSIGHTS =====

1. Content Mix:

Netflix contains 6131 Movies and 2676 TV Shows in the dataset.

2. Most Common Genres:

The most common genres are:

|                        |      |
|------------------------|------|
| International Movies   | 2752 |
| Dramas                 | 2427 |
| Comedies               | 1674 |
| International TV Shows | 1351 |
| Documentaries          | 869  |

Name: count, dtype: int64

3. Leading Content-Producing Countries:

The leading countries by number of titles are:

|                |      |
|----------------|------|
| United States  | 3689 |
| India          | 1046 |
| Unknown        | 831  |
| United Kingdom | 804  |
| Canada         | 445  |

Name: count, dtype: int64

4. Content Growth:

The yearly content trend shows how Netflix's content library has changed over time.

5. Content Ratings:

The rating distribution shows which audience categories are most represented in the dataset.

**9. Present the complete data story using presentation slides, Tableau, Power BI, or a Jupyter Notebook.**

**Cell 23**

```

print("""
===== ACTIONABLE RECOMMENDATIONS =====
1. Content Strategy:
    Netflix can monitor the distribution of Movies and TV Shows
    to maintain a balanced content portfolio.
2. Genre Strategy:
    Popular genres can be monitored to understand viewer interests
    and guide future content planning.
3. Regional Strategy:
    Countries with strong content representation can be analyzed
    for regional content opportunities.
4. Audience Strategy:
    Content ratings can be used to understand the target audience
    distribution and support appropriate content planning.
5. Trend Monitoring:
    Year-wise content trends can be continuously monitored to
    understand changes in content production and availability.
6. Performance Monitoring:
    KPIs and dashboards can be updated regularly so that analysts
    and managers can monitor changes in the content library.
""")

```

===== ACTIONABLE RECOMMENDATIONS =====

1. Content Strategy:  
Netflix can monitor the distribution of Movies and TV Shows to maintain a balanced content portfolio.
2. Genre Strategy:  
Popular genres can be monitored to understand viewer interests and guide future content planning.
3. Regional Strategy:  
Countries with strong content representation can be analyzed for regional content opportunities.
4. Audience Strategy:

Content ratings can be used to understand the target audience distribution and support appropriate content planning.

5. Trend Monitoring:

Year-wise content trends can be continuously monitored to understand changes in content production and availability.

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## QUESTION-ANSWERS

1. What is data storytelling? How is it different from data visualization?

Answer: Data storytelling combines data, visualizations, and a narrative to explain insights. Data visualization mainly presents data graphically, while storytelling explains the meaning and importance of the data.

2. Why is storytelling important in business analytics and decision-making?

Answer: Storytelling makes complex data easier to understand and helps stakeholders identify important insights, trends, and problems for better decision-making.

3. What are the essential components of an effective data story?

Answer: The main components are data, visualizations, insights, narrative, and recommendations.

4. How do Key Performance Indicators (KPIs) enhance business reporting?

Answer: KPIs provide measurable values that help organizations monitor performance, compare results, and identify areas that need improvement.

5. Why should visualizations be arranged in a logical sequence while presenting analytical findings?

Answer: A logical sequence creates a clear story from the problem → analysis → findings → recommendations, making the information easier to understand.

6. What factors should be considered while selecting visualizations for a business presentation?

Answer: Consider the type of data, purpose of analysis, audience, readability, accuracy, and simplicity when selecting visualizations.

7. Explain how dashboards and storytelling complement each other in Business Intelligence.

Answer: Dashboards provide interactive visual information, while storytelling explains the meaning of that information. Together, they help stakeholders understand data and make informed decisions.

8. What challenges may arise while communicating analytical insights to non-technical stakeholders?

Answer: Common challenges include technical terminology, complex visualizations, large amounts of data, and difficulty understanding statistical concepts.

9. Give two real-world examples where data storytelling has influenced business or policy decisions.

Answer:Netflix: Content viewing and user data can be used to understand audience preferences and guide content decisions. Public health: COVID-19 dashboards and visualizations have helped communicate infection trends and support public-health decisions.

10. How can effective data storytelling improve strategic planning and organizational performance?

Answer: Data storytelling helps organizations understand trends, identify opportunities and risks, communicate findings clearly, and make better strategic decisions.